

Consider the data set and perform the Apriori Algorithm and FP algorithm support:3 and confidence=50%

Customer ID	Transaction ID	Items Bought
1	0001	{a, d, e}
1	0024	{a, b, c, e}
2	0012	{a, b, d, e}
2	0031	{a, c, d, e}
3	0015	{b, c, e}
3	0022	{b, d, e}
4	0029	{c, d}
4	0040	{a, b, c}
5	0033	{a, d, e}
5	0038	{a, b, e}

Consider the data set and perform the Apriori Algorithm and FP algorithm support:3 and confidence=50%

Consider the market basket transactions shown in the above table.

- (a) What is the maximum number of association rules that can be extracted from this data (including rules that have zero support)?
- (b) What is the maximum size of frequent itemsets that can be extracted (assuming minsup > 0)?

Transaction ID	Items Bought
1	{Milk, Beer, Diapers}
2	{Bread, Butter, Milk}
3	{Milk, Diapers, Cookies}
4	{Bread, Butter, Cookies}
5	{Beer, Cookies, Diapers}
6	{Milk, Diapers, Bread, Butter}
7	{Bread, Butter, Diapers}
8	{Beer, Diapers}
9	{Milk, Diapers, Bread, Butter}
10	{Beer, Cookies}

Bayes classification and descion tree (using training and test data)

RID	age	income	student	credit_rating	Class: buys_computer
1	<=30	high	no	fair	no
2	<=30	high	no	excellent	no
3	31 . . . 40	high	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	31 . . . 40	low	yes	excellent	yes
8	<=30	medium	no	fair	no
9	<=30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<=30	medium	yes	excellent	yes
12	31 . . . 40	medium	no	excellent	yes
13	31 . . . 40	high	yes	fair	yes
14	>40	medium	no	excellent	no

Analysis the dataset “diabetes. csv” how the diabetes trend is for different age people, using linear regression and multiple regression.

Implement using WEKA for the given Suppose a database has five transactions. Let min sup= 50%(2) and min con f = 80%.

Transactions	Items
T1	(M, O, N, K, E, Y)
T2	(D, O, N, K, E, Y)
T3	(M, A, K, E)
T4	(M, U, C, K, Y)
T5	(C,O, O, K, I ,E)

- Find all frequent item sets using Apriori algorithm
- Also draw FP-Growth Tree

Prediction of Categorical Data using Decision Tree Algorithm through WEKA using any datasets. a) Tree b) Preprocess c) Logistic

Create the dataset using ARFF file format:

Transaction ID	Items
T1	Hot Dogs, Buns, Ketchup
T2	Hot Dogs, Buns
T3	Hot Dogs, Coke, Chips
T4	Chips, Coke
T5	Chips, Ketchup
T6	Hot Dogs, Coke, Chips

a.Find the **frequent itemsets** and generate **association rules** on this. Assume that minimum support threshold ($s = 33.33\%$) and minimum confident threshold ($c = 60\%$).

b.List the various rule generated by apriori and FP tree algorithm ,mention wheather accepted or rejcted.

Prediction of Categorical Data using Rule base classification and decision tree classification through WEKA using any datasets. Compare the accuracy using two algorithm and plot the graph

APRIORI

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator
Choose **Apriori** -N 10 -T 0 -C 0.5 -D 0.05 -U 1.0 -M 0.3 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)
11:24:01 - Apriori

Associator output

```

Scheme: weka.associations.Apriori -N 10 -T 0 -C 0.5 -D 0.05 -U 1.0 -M 0.3 -S -1.0 -c -1
Relation: market_basket
Instances: 10
Attributes: 5
a
b
c
d
e

=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.45 (4 instances)
Minimum metric <confidence>: 0.5
Number of cycles performed: 11

Generated sets of large itemsets:

Size of set of large itemsets L(1): 8

Size of set of large itemsets L(2): 10

Size of set of large itemsets L(3): 3

Best rules found:

1. c=>0 5 ==> e=1 5 <conf:(1)> lift:(1.25) lev:(0.1) [0] conv:(1)
2. b=>0 4 ==> d=1 4 <conf:(1)> lift:(1.67) lev:(0.16) [1] conv:(1.6)
3. d=>0 4 ==> b=1 4 <conf:(1)> lift:(1.67) lev:(0.16) [1] conv:(1.6)
4. a=1 c=>0 4 ==> e=1 4 <conf:(1)> lift:(1.25) lev:(0.08) [0] conv:(0.8)
5. a=1 d=1 4 ==> e=1 4 <conf:(1)> lift:(1.25) lev:(0.08) [0] conv:(0.8)
6. c=>0 d=1 4 ==> e=1 4 <conf:(1)> lift:(1.25) lev:(0.08) [0] conv:(0.8)
7. a=1 7 ==> e=1 6 <conf:(0.86)> lift:(1.07) lev:(0.04) [0] conv:(0.7)
8. b=1 6 ==> e=1 5 <conf:(0.83)> lift:(1.04) lev:(0.02) [0] conv:(0.6)
9. d=1 6 ==> e=1 5 <conf:(0.83)> lift:(1.04) lev:(0.02) [0] conv:(0.6)

```

Status
OK

Log

FP GROWTH

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator
Choose **FPGrowth** -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1

Start Stop

Result list (right-click for ...)
11:24:01 - Apriori
11:26:27 - FPGrowth

Associator output

```

=== Run information ===

Scheme: weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1
Relation: market_basket
Instances: 10
Attributes: 5
a
b
c
d
e

=== Associator model (full training set) ===

FPGrowth found 5 rules (displaying top 5)

1. [a=1, d=1]: 4 ==> [e=1]: 4 <conf:(1)> lift:(1.25) lev:(0.08) conv:(0.8)
2. [d=1, b=1]: 2 ==> [e=1]: 2 <conf:(1)> lift:(1.25) lev:(0.04) conv:(0.4)
3. [a=1, d=1, b=1]: 1 ==> [e=1]: 1 <conf:(1)> lift:(1.25) lev:(0.02) conv:(0.2)
4. [a=1, d=1, c=1]: 1 ==> [e=1]: 1 <conf:(1)> lift:(1.43) lev:(0.03) conv:(0.3)
5. [a=1, d=1, c=1]: 1 ==> [e=1]: 1 <conf:(1)> lift:(1.25) lev:(0.02) conv:(0.2)

```

Status
OK

Log

31°C Mostly sunny

Search

ENG IN

11:26 21-08-2026

NB

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes**

Test options

- ☒ Use training set
- ☐ Supplied test set
- ☐ Cross-validation Folds 10
- ☐ Percentage split % 66
-

(Norm) buys_computer

Result list (right-click for options)

- 11:29:58 - bayes.NaiveBayes

Classifier output

```
Time taken to build model: 0 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances      13           92.8571 %
Incorrectly Classified Instances    1           7.1429 %
Kappa statistic                    0.8372
Mean absolute error                 0.2917
Root mean squared error             0.3392
Relative absolute error             62.8233 %
Root relative squared error         70.7422 %
Total Number of Instances          14

=== Detailed Accuracy By Class ===

              TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
              1.000    0.200    0.900     1.000    0.947     0.849    0.922    0.947    yes
              0.800    0.000    1.000     0.800    0.889     0.849    0.911    0.911    no
Weighted Avg.   0.929    0.129    0.936     0.929    0.926     0.849    0.918    0.934

=== Confusion Matrix ===

a b  <-- classified as
9 0 | a = yes
1 4 | b = no
```

Status: OK x0

J48

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options

- ☒ Use training set
- ☐ Supplied test set
- ☐ Cross-validation Folds 10
- ☐ Percentage split % 66
-

(Norm) buys_computer

Result list (right-click for options)

- 11:29:58 - bayes.NaiveBayes
- 11:31:11 - trees.J48

Classifier output

```
=== Run information ===

Scheme:      weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:    buys_computer
Instances:    14
Attributes:   5
  age
  income
  student
  credit_rating
  buys_computer
Test mode:    evaluate on training data

=== Classifier model (full training set) ===

J48 pruned tree
-----

age <= 30
| student = yes: yes (2.0)
| student = no: no (3.0)
age = 31..40: yes (4.0)
age > 40
| credit_rating = fair: yes (3.0)
| credit_rating = excellent: no (2.0)

Number of Leaves :    5
Size of the tree :    8

Time taken to build model: 0.01 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds
```

Status: OK x0

Time taken to build model: 0.01 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances	14	100	%
Incorrectly Classified Instances	0	0	%
Kappa statistic	1		
Mean absolute error	0		
Root mean squared error	0		
Relative absolute error	0	%	
Root relative squared error	0	%	
Total Number of Instances	14		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	yes
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	no
Weighted Avg.	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	

=== Confusion Matrix ===

a b <-- classified as
9 0 | a = yes
0 5 | b = no

Lr:

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The 'Classifier' dropdown is set to 'LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4'. The 'Test options' section has 'Use training set' selected. The 'Classifier output' pane displays the following information:

=== Run information ===

Scheme: weka.classifiers.functions.LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4
Relation: diabetes-weka.filters.unsupervised.attribute.Remove-R1-7
Instances: 768
Attributes: 2
Age
Outcome
Test mode: evaluate on training data

=== Classifier model (full training set) ===

Linear Regression Model

Outcome =

$$0.0097 * \text{Age} + 0.0276$$

Time taken to build model: 0.1 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0.01 seconds

=== Summary ===

Correlation coefficient	0.2384
Mean absolute error	0.4286
Root mean squared error	0.4629
Relative absolute error	94.3186 %
Root relative squared error	97.1178 %
Total Number of Instances	768

The 'Result list' on the left shows '11:47:37 - functions.LinearRegression' selected. The status bar at the bottom indicates 'Status OK'.

Multiple Regression

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier

Choose **LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4**

Test options

☒ Use training set

☐ Supplied test set

☐ Cross-validation Folds

☐ Percentage split %

(Num) Outcome

Result list (right-click for options)

11:49:24 - functions.LinearRegression

Classifier output

Outcome

Test mode: evaluate on training data

=== Classifier model (full training set) ===

Linear Regression Model

Outcome =

0.0209 * Pregnancies +
 0.0057 * Glucose +
 -0.0024 * BloodPressure +
 0.0131 * BMI +
 0.1403 * DiabetesPedigreeFunction +
 0.0028 * Age +
 -0.8363

Time taken to build model: 0.03 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0.01 seconds

=== Summary ===

Correlation coefficient	0.5494
Mean absolute error	0.3326
Root mean squared error	0.3983
Relative absolute error	73.1929 %
Root relative squared error	83.5591 %
Total Number of Instances	768

Status

OK

TREE

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier

Choose **J48 -C 0.25 -M 2**

Test options

☒ Use training set

☐ Supplied test set

☐ Cross-validation Folds

☐ Percentage split %

(Nom) Outcome

Result list (right-click for options)

11:58:40 - trees.J48

Classifier output

=== Run information ===

Scheme: weka.classifiers.trees.J48 -C 0.25 -M 2
 Relation: diabetes-weka.filters.unsupervised.attribute.NumericToNominal-89
 Instances: 768
 Attributes: 9
 Pregnancies
 Glucose
 BloodPressure
 SkinThickness
 Insulin
 BMI
 DiabetesPedigreeFunction
 Age
 Outcome

Test mode: evaluate on training data

=== Classifier model (full training set) ===

J48 pruned tree

```

=====
Glucose <= 127
| BMI <= 26.4: 0 (132.0/3.0)
| BMI > 26.4
| | Age <= 28: 0 (180.0/22.0)
| | Age > 28
| | | Glucose <= 99: 0 (55.0/10.0)
| | | Glucose > 99
| | | | DiabetesPedigreeFunction <= 0.56: 0 (84.0/34.0)
| | | | DiabetesPedigreeFunction > 0.56
| | | | | Pregnancies <= 6
| | | | | Age <= 30: 1 (4.0)
| | | | | Age > 30
| | | | | Age <= 34: 0 (7.0/1.0)
| | | | | Age > 34
| | | | | BMI <= 33.1: 1 (6.0)

```

Status

Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options:
☒ Use training set
☐ Supplied test set
☐ Cross-validation Folds: 10
☐ Percentage split %: 66
 More options...

(Nom) Outcome: Start Stop

Result list (right-click for options):
 11:58:40 - trees.J48

Classifier output:

```

| | | | | | | | BMI <= 33.1: 1 (6.0)
| | | | | | | | BMI > 33.1: 0 (4.0/1.0)
| | | | | | | | Pregnancies > 6: 1 (13.0)
| | | | | | | | Glucose > 127
| | | | | | | | BMI <= 29.9
| | | | | | | | Glucose <= 145: 0 (41.0/6.0)
| | | | | | | | Glucose > 145
| | | | | | | | Age <= 25: 0 (4.0)
| | | | | | | | Age > 25
| | | | | | | | Age <= 61
| | | | | | | | BMI <= 27.1: 1 (12.0/1.0)
| | | | | | | | BMI > 27.1
| | | | | | | | BloodPressure <= 82
| | | | | | | | DiabetesPedigreeFunction <= 0.396: 1 (0.0/1.0)
| | | | | | | | DiabetesPedigreeFunction > 0.396: 0 (3.0)
| | | | | | | | BloodPressure > 82: 0 (4.0)
| | | | | | | | Age > 61: 0 (4.0)
| | | | | | | | BMI > 29.9
| | | | | | | | Glucose <= 157
| | | | | | | | BloodPressure <= 61: 1 (15.0/1.0)
| | | | | | | | BloodPressure > 61
| | | | | | | | Age <= 30: 0 (40.0/13.0)
| | | | | | | | Age > 30: 1 (60.0/17.0)
| | | | | | | | Glucose > 157: 1 (52.0/12.0)

Number of Leaves :    20
Size of the tree :    39

Time taken to build model: 0.06 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0.01 seconds

=== Summary ===

```

Status: OK Log x0

Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options:
☒ Use training set
☐ Supplied test set
☐ Cross-validation Folds: 10
☐ Percentage split %: 66
 More options...

(Nom) Outcome: Start Stop

Result list (right-click for options):
 11:58:40 - trees.J48

Classifier output:

```

| | | | | | | | Glucose > 127: 1 (52.0/12.0)
| | | | | | | |
Number of Leaves :    20
Size of the tree :    39

Time taken to build model: 0.06 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0.01 seconds

=== Summary ===

Correctly Classified Instances      646      84.1146 %
Incorrectly Classified Instances    122      15.8854 %
Kappa statistic                    0.6319
Mean absolute error                 0.2383
Root mean squared error             0.3452
Relative absolute error             52.4339 %
Root relative squared error         72.4207 %
Total Number of Instances          768

=== Detailed Accuracy By Class ===

      TP Rate  FP Rate  Precision  Recall   F-Measure  MCC     ROC Area  PRC Area  Class
      0.936   0.336   0.839    0.936   0.885     0.642   0.888    0.915     0
      0.664   0.064   0.848    0.664   0.745     0.642   0.888    0.808     1
Weighted Avg.   0.841   0.241   0.842    0.841   0.836     0.642   0.888    0.878

=== Confusion Matrix ===

  a  b  <-- classified as
468 32 | a = 0
 90 178 | b = 1

```

Status: OK Log x0

35°C Search ENG dx 1400